

Designing Building Blocks for Stacked Polyphase Bridge Converters in High-Voltage Traction Inverters

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Abstract—Stacked polyphase bridge (SPB) converters divide a high traction-battery voltage among series-connected low-voltage three-phase submodules, which makes 100 V gallium-nitride (GaN) transistors usable in 800 V drives. This paper presents the design of the phase-leg building block that such a converter is assembled from: a 75 V, 100 A_{rms} half bridge with two paralleled 100 V GaN transistors per switch on a 30 × 28 mm four-layer board. Two mirrored commutation cells with vertical power loops and a centre gate driver give 0.374 nH per cell, matched 1.7 nH gate loops and a predicted dynamic current imbalance below 0.1 %. The design is carried out with a model chain in which one parametric geometry drives the printed-circuit-board (PCB) placement, the Ansys Q3D extraction, LTspice switching simulation, a frequency-domain copper-loss model and an Ansys Icepak thermal model. Three results are of general interest: the switching-frequency harmonics of the dc-link current add one third to the dc copper loss of the board, the classical overlap loss formula is valid for a GaN turn-off only above a critical gate resistance that is derived and verified, and the board copper loss rather than the devices sets the thermal design. At 50 kHz the block reaches 99.53 % efficiency at 2.65 kW per leg with a junction temperature of 72 °C on a 60 °C cold plate.

Index Terms—Gallium nitride, paralleling, parasitic inductance, PCB layout, stacked polyphase bridge, switching loss, thermal design, traction inverter.

I. INTRODUCTION

Raising the battery voltage of electric vehicles to 800 V reduces the current and the charging time, but it forces the traction inverter to 1200 V devices, steep voltage steps at the machine terminals, and bulky dc-link capacitors when a two-level topology is used. The stacked polyphase bridge (SPB) converter [1], [2] offers an alternative: N conventional three-phase bridges are connected in series on the dc side, each with its own dc-link capacitor and each feeding one three-phase winding set of a multiphase machine. Every submodule sees only V_{dc}/N , so low-voltage transistors with a far better on-resistance and switching figure of merit can be used, the submodule capacitors shrink because the switching frequency can be raised [3], the voltage step at the windings is reduced [9], and a faulty submodule can be bypassed [3]. Analysis of the dc-link stability [5], control [2], [9], [10], machine design [7], [8] and an integrated prototype [6] have been reported, a comparative evaluation for integrated motor drives found the SPB with 650 V GaN transistors competitive

with a three-level flying-capacitor inverter [20], and a GaN-based SPB was shown to reduce the switching losses from 3.2 % to 0.5 % of the output power compared with silicon MOSFETs at 100 kHz [4].

The building block of an SPB, however, is a demanding power stage in its own right. At a submodule voltage of 75 V and a phase current of 100 A_{rms}, a single GaN die is not sufficient, and paralleled GaN half bridges are known to fail by drain-source overshoot [11], by uneven dynamic current sharing caused by asymmetric loops [12], [13] and by gate-voltage disturbances [13], [14]. Low-voltage paralleled-GaN inverters have been demonstrated with symmetric “butterfly” layouts [15], eight parallel devices on a symmetric gate board [16] and hybrid insulated-metal-substrate packaging [13]; efficiency-density co-optimization has been formalized for a 100 kW GaN inverter [17].

This paper addresses the design of the SPB building block as a whole. Its contributions are:

- 1) a voltage-class selection showing that, at a given bus voltage and phase current, the minimum semiconductor loss of an SPB scales with the device time constant $R_{DS(on)}C_{oss,tr}$, which favors 100 V GaN transistors;
- 2) a symmetric two-cell, centre-driver layout for two paralleled 100 V GaN transistors per switch, and a model chain that keeps the PCB placement, the electromagnetic and thermal models and the circuit simulation consistent;
- 3) a frequency-domain board copper-loss model built on a Q3D frequency sweep, which shows that the switching harmonics of the dc-link current add about one third to the dc copper loss;
- 4) a criterion for when the overlap loss formula $V^2 I / (2 dv/dt)$ applies to the turn-off of GaN transistors, verified over six gate-resistor pairs, and the cost of slow switching;
- 5) a conduction thermal model that identifies the board copper loss, rather than the devices, as the limiting thermal path.

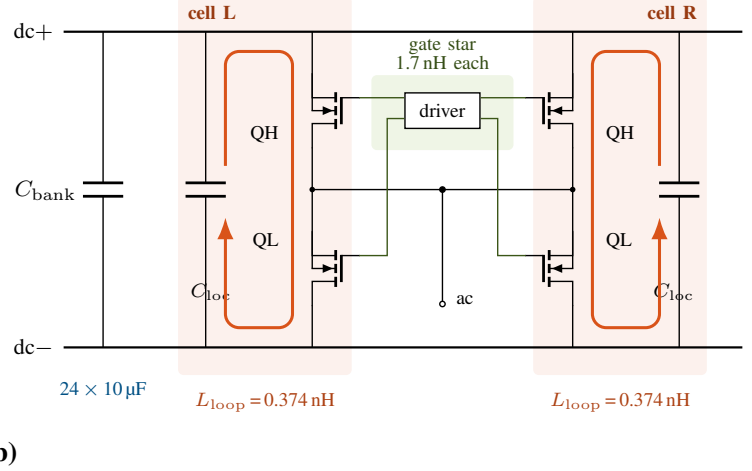
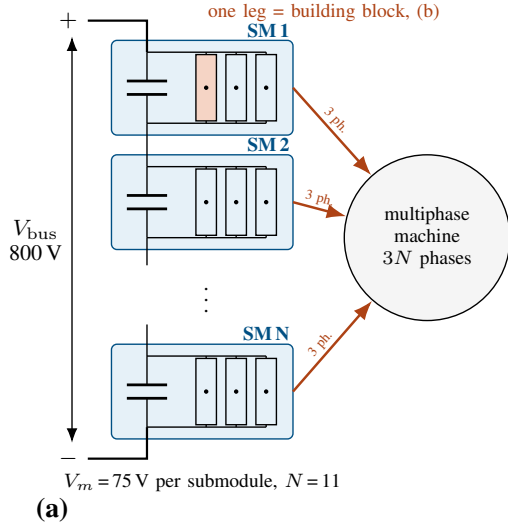


Fig. 1. (a) Stacked polyphase bridge (SPB) converter: N three-phase submodules (SM) are connected in series on the dc side and feed the N three-phase winding sets of a multiphase machine; each submodule operates at $V_m = V_{\text{bus}}/N$. (b) Building block designed in this paper, i.e., one phase leg of a submodule: two mirror-symmetric commutation cells (red) with one high-side (QH) and one low-side (QL) EPC2361 each, so that two transistors are paralleled per switch, local MLCCs C_{loc} in every cell, a common MLCC bank C_{bank} and a centre gate driver (green) with equal gate loops. The red arrows indicate the commutation loop of each cell (Q3D, including capacitor ESL).

II. SPB CONVERTER AND BUILDING-BLOCK SPECIFICATION

A. Specification

For an 800 V battery, eleven submodules of 75 V each leave a 15 V design margin up to 90 V and 25 V to the 100 V rating of the transistors for overshoot, ringing and battery-voltage variation. With 100 A_{rms} per phase ($\bar{I} = 141$ A) and sinusoidal pulse-width modulation, one leg delivers 2.65 kW at a modulation index of one and 3.05 kW with space-vector modulation ($M = 1.15$). The design targets are a leg efficiency of 99.5 % (13.3 W), a peak V_{DS} below 90 V at a 13 % overcurrent (160 A), an off-state gate bump below 1 V and a nominal switching frequency of 50 kHz. Using the device power utilization of [16], $\text{DPU} = V_{\text{dc}} I_{\text{ph}} R_{\text{DS(on)}} / (n V_{\text{DS}}^2) \cdot 10^4$, the block is rated at a DPU of 3.75, comparable to the continuously operated paralleled-GaN inverters reviewed in [16] (2.3–9.5).

B. Choice of the voltage class

The submodule voltage, and with it the transistor voltage class, is the first design decision of an SPB. When a bus voltage V_{bus} is split into $N = V_{\text{bus}}/V_m$ submodules that share the power equally, every submodule carries the same phase current I , independent of N . With n transistors per switch, the conduction loss of all submodules scales with $N I^2 R_{\text{DS(on)}}/n$ and the loss associated with the output charge with $N n f_{\text{sw}} Q_{\text{oss}} V_m$. With $Q_{\text{oss}} \approx C_{\text{oss,tr}} V_m$ and an optimal number of devices, $n^* = I \sqrt{R_{\text{DS(on)}} / (f_{\text{sw}} Q_{\text{oss}} V_m)}$, the sum of the two over all three legs of all submodules becomes

$$P_{\text{min}} = 6 I V_{\text{bus}} \sqrt{f_{\text{sw}} R_{\text{DS(on)}} C_{\text{oss,tr}}}. \quad (1)$$

To first order the voltage class cancels, and candidates are ranked by the device time constant $\tau = R_{\text{DS(on)}} C_{\text{oss,tr}}$.

TABLE I
BUILDING-BLOCK SPECIFICATION AND KEY RESULTS

Specification	
Submodule voltage / phase current	75 V / 100 A _{rms} (141 A peak)
Power per leg ($M = 1 / 1.15$)	2.65 / 3.05 kW
Targets	$\eta \geq 99.5\%$, $\hat{V}_{\text{DS}} < 90$ V at 160 A
Design	
Transistors	2 × EPC2361 (100 V, 0.75 mΩ) per switch
Board	30 × 28.2 mm, 4 layers, 2 oz, 0.1 mm L1–L2
DC link per leg	24 × 10 μF + 12 × 1 μF MLCC
$R_{\text{g,on}}/R_{\text{g,off}}$	2/0 Ω
Results (simulation)	
Commutation loop per cell (Q3D)	0.312 nH copper, 0.374 nH with ESL
Gate loop per transistor (Q3D)	1.70 nH, equal for all four
Peak V_{DS} at 160 A (HS/LS)	83.1 / 85.1 V
Loss / efficiency at 50 kHz	12.49 W / 99.53 %
T_j at 60 °C cold plate	72 °C (67 °C with bottom pad)

Fig. 2 and Table II compare ten EPC and Infineon devices from the 100, 200, 350 and 650 V classes, each at a submodule voltage of about 75 % of its rating (60 % for 650 V). The 100 V devices reach $\tau = 1.35$ –2.4 ps, the 200 V devices 4.1–5.8 ps, the only available 350 V device 6.9 ps and the 650 V devices about 4.9 ps. An SPB built from EPC2361 therefore has a 1.8 times lower minimum loss than one built from the best 200 V device and a 1.9 times lower one than with 650 V devices. Three practical arguments point the same way. First, at 100 A per submodule the optimum number of high-voltage devices per switch becomes impractical (12 for IGT65R025D2), whereas EPC2361 needs about 3.8; the overlap and copper losses, not included in (1), reduce this to two (Section II-C). Second, the 3 × 5 mm packages of the 100 V class allow sub-nanohenry commutation loops (Section III). Third, the on-resistance of the 650 V devices rises by a factor of 2.1 at 150 °C, against

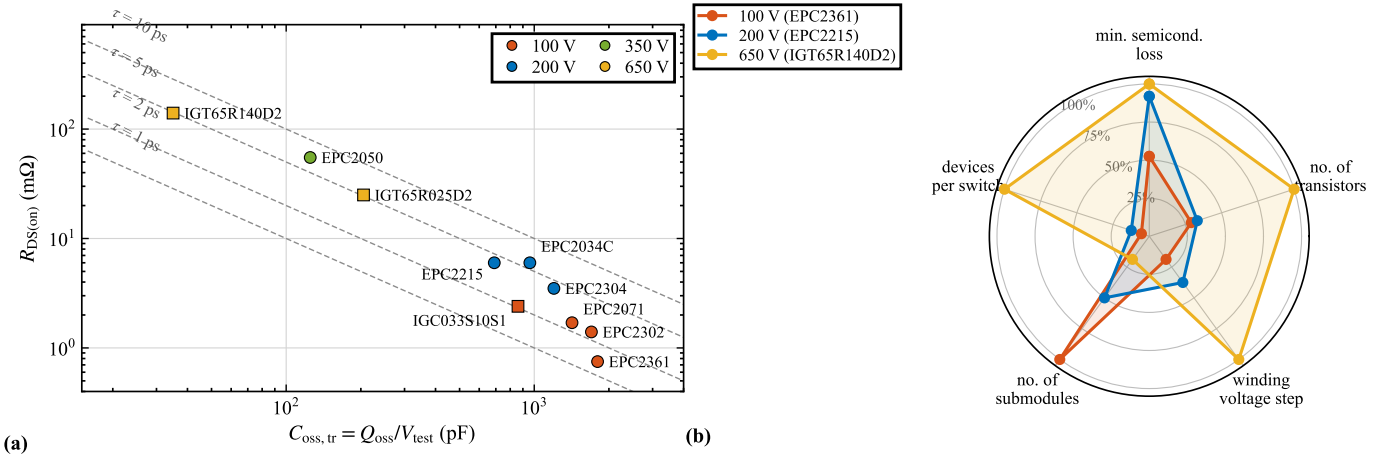


Fig. 2. Voltage-class selection for an 800 V SPB with 100 A_{rms} phase current at 50 kHz. (a) Typical on-resistance at 25 °C against the time-related output capacitance $C_{oss,tr} = Q_{oss}/V_{test}$ of off-the-shelf GaN transistors (circles: EPC, squares: Infineon; datasheet values), with lines of constant device time constant $\tau = R_{DS(on)}C_{oss,tr}$. (b) Comparison of SPBs built from the best device of the 100, 200 and 650 V classes, each axis normalized to its maximum: minimum conduction plus output-charge loss (1), optimum number of devices per switch n^* , number of submodules N , voltage step at the winding V_m and total number of transistors $6Nn^*$.

TABLE II
OFF-THE-SHELF GAN TRANSISTORS FOR AN 800 V SPB (100 A_{rms}, 50 kHz)

Device	V_{DS} [V]	$R_{DS(on)}$ [mΩ]	$C_{oss,tr}$ [pF]	V_m [V]	τ [ps]	N	P_{min} rel.
EPC2361	100	0.75	1800	75	1.35	10.7	1.0
IGC033S10S1	100	2.4	860	75	2.06	10.7	1.2
EPC2302	100	1.4	1700	75	2.38	10.7	1.3
EPC2071	100	1.7	1420	75	2.41	10.7	1.3
EPC2215	200	6	690	150	4.14	5.3	1.8
EPC2304	200	3.5	1200	150	4.20	5.3	1.8
EPC2034C	200	6	960	150	5.76	5.3	2.1
EPC2050	350	55	125	260	6.88	3.1	2.3
IGT65R140D2	650	140	35	400	4.90	2.0	1.9
IGT65R025D2	650	25	205	400	5.12	2.0	1.9

1.8 for EPC2361. The price is a larger number of submodules (11 instead of 2), i.e., more gate drivers, auxiliary supplies and control channels, which the modularity of the SPB accommodates [3]. This contrasts with the two-submodule, 650 V GaN SPB of [20], where the submodule count rather than the device figure of merit was minimized. EPC2361 is therefore selected, and the remainder of this paper designs the 100 V building block around it.

C. Number of devices in parallel

The number of EPC2361 per switch follows from the trade-off between conduction loss ($\propto 1/n$) and the current-independent part of the switching energy, which grows with n . One device per switch misses the efficiency target (17.5 W, 6.4 W per device), whereas three and four devices reduce the loss by only 1.3–1.7 W at the cost of longer loops and a larger sharing risk; two devices per switch are used.

III. LAYOUT AND PARASITIC DESIGN

A. Symmetric two-cell layout

The block is split into two mirror-symmetric commutation cells at $x = \pm 7.5$ mm (Fig. 3). Each cell carries one high-side and one low-side transistor and six 1 μ F 0805 capacitors at the high-side drain. The commutation loop is vertical: the forward path on layer L1 is closed on the dc– plane on L2, 0.1 mm below, so that the opposing currents cancel their flux. The dc+ plane on L3 and the ac plane on L4 carry the load current to the terminals on the symmetry line, and a bank of 24 10 μ F 1210 capacitors above the cells feeds both cells equally, which avoids the shared, cascaded loop inductance identified in [11]. The gate driver sits in the 6 mm corridor between the cells and reaches all four transistors through equal gate stars with Kelvin source returns. The board measures 30 $\times$ 28.2 mm (8.5 cm², 0.31 kW/cm² at board level).

The same parametric geometry generates the KiCad placement (Fig. 3a), the Q3D model (Fig. 3b) and the Icepak model (Fig. 3c), so that a change of, e.g., the prepreg thickness propagates to the inductances, losses and temperatures without manual steps.

B. Power and gate loop

The partial resistance and inductance matrix of the copper is extracted with Ansys Q3D at 100 MHz, every component being represented by source and sink terminals on its pads, and is exported to LTspice as coupled inductors. Exciting both cells simultaneously gives a commutation-loop inductance of 0.312 nH (copper) and 0.374 nH including the capacitor ESL per cell. Each transistor sees this loop; paralleling does not reduce the inductance, but each cell commutates half of the load current, so that the di/dt per loop and thus the overshoot are halved. The value excludes the package inductance and routing details and is therefore a lower bound.

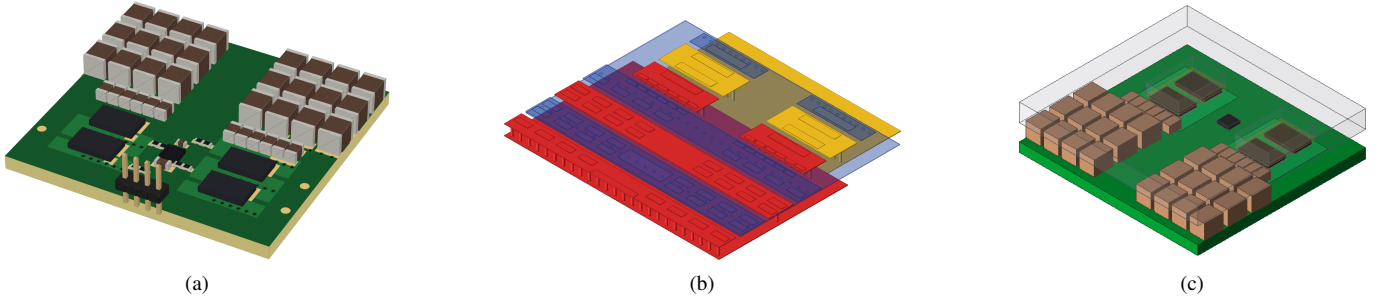


Fig. 3. One parametric geometry, three models. (a) KiCad placement with the component 3D models (30×28.2 mm, four layers, 2 oz copper): MLCC bank (top), local MLCCs, the two mirrored cells with QH and QL, and the centre driver. (b) Ansys Q3D copper model with the dc+ (red), dc- (blue, transparent) and ac (yellow) nets and the via rows of the vertical loops. (c) Ansys Icepak model with gap pads, pedestals and the aluminium cold plate (shown above the board).

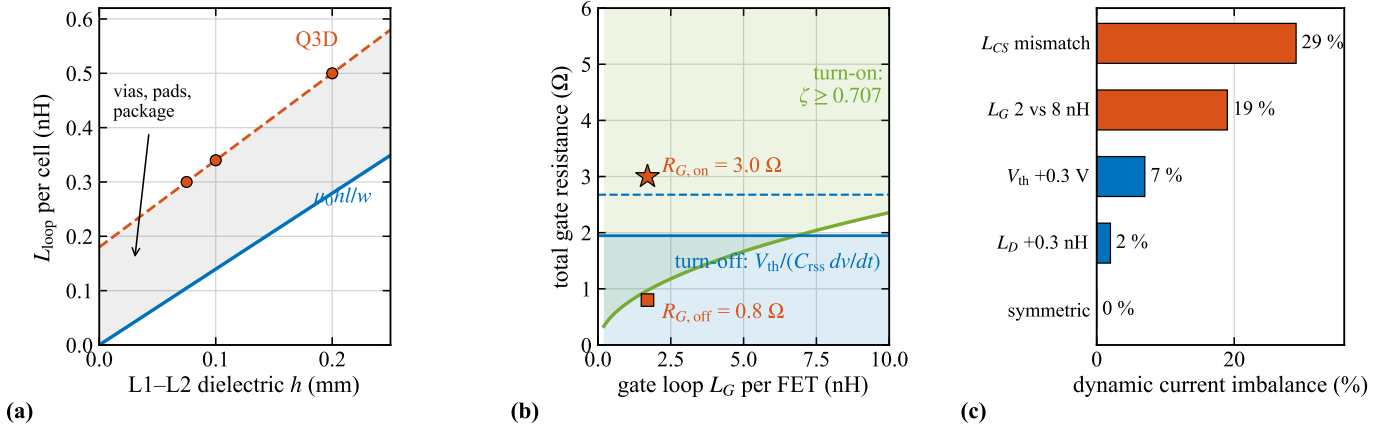


Fig. 4. Parasitic design of the building block. (a) Commutation-loop inductance per cell against the L1–L2 dielectric thickness h : Q3D extraction (circles, linear fit dashed) and parallel-plate estimate $\mu_0 h l / w$; the shaded difference is the fixed contribution of vias, pads and package. (b) Gate-resistor design window per transistor: lower bound for critical damping of the gate loop at turn-on (green, $\zeta \geq 0.707$) and upper bound against Miller turn-on at turn-off (blue, solid for $V_{th,min} = 0.8$ V, dashed for 1.1 V), with the selected resistances at $L_G = 1.7$ nH. (c) Dynamic current imbalance of two paralleled EPC2361 (75 V, 120 A, LTspice) for one asymmetry at a time: common-source inductance mismatch without Kelvin connection, unequal gate loops, threshold-voltage spread and drain-path mismatch.

The dielectric thickness h between L1 and L2 is the dominant geometric parameter. Q3D gives 1.60 nH per millimetre of dielectric, close to the 1.40 nH/mm of the parallel-plate estimate $\mu_0 h l / w$ used in [13], [16] (Fig. 4a), plus a fixed 0.18 nH from vias, pads and package that no dielectric reduction can remove. At 0.1 mm (two-ply prepreg) the fixed part is already half of the loop.

The gate star gives 1.70 nH per transistor. A side-mounted driver, as in a conventional single-row layout, would give 2.0 and 7.7 nH to the near and far transistors (Q3D); the centre driver makes both branches equal by construction.

C. Gate resistors and current sharing

Treating the gate loop as a series RLC with C_{iss} gives a lower bound for the turn-on resistance, $R_{G,on} \geq 2\zeta \sqrt{L_G / C_{iss}}$, and the Miller current an upper bound for the turn-off resistance, $R_{G,off} \leq V_{th} / (C_{rss} dv/dt)$ [13] (Fig. 4b). Inside this window, an LTspice sweep with the extracted parasitics at 160 A selects $R_{G,on} / R_{G,off} = 2/0 \Omega$: below 2 Ω the high-side overshoot exceeds 90 V (112 V at 0.5 Ω), above it

E_{on} rises. The peak V_{DS} is 83.1 V and 85.1 V for the high and low side, and the switching speeds are 12 and 32 V/ns at turn-on and turn-off. The off-state bump of the high-side gate is 0.98 V; it exceeds the minimum threshold of 0.8 V and is the tightest margin of the design.

A two-transistor sensitivity study (Fig. 4c) ranks the asymmetries: a common-source mismatch without Kelvin returns (29%) and a gate-loop mismatch of 2 versus 8 nH (19%, 4:1 split of E_{on}) dominate, a threshold-voltage difference of 0.3 V gives 7%, and a drain-path mismatch of 0.3 nH only 2%. The gate-loop case explains the centre driver: a mismatch ΔL_G delays one gate by about $\Delta L_G / R_{tot} \approx 2$ ns, half the 4 ns current transition. In the complete building block the two transistors carry 78.53 and 78.63 A at turn-off.

IV. SWITCHING ENERGIES AND SLOW SWITCHING

With the selected resistors, one switch position dissipates $E_{on} = 43 \mu J$ and $E_{off} = 8.1 \mu J$ at 141 A. The turn-on energy consists of a current-independent part of 12 μJ , of the order of $Q_{oss} V$, and an overlap part that follows $V^2 I / (2 dv/dt)$ within

TABLE III
GATE-RESISTOR PAIRS AT 139 A AND LEG EFFICIENCY AT 50 kHz

$R_{g,on}/R_{g,off}$ [Ω]	E_{on} [μJ]	E_{off} [μJ]	dv/dt on/off [V/ns]	HS bump [V]	P_{sw} [W]	η [%]
2/0 (selected)	40.9	8.0	13/27	0.99	1.87	99.53
2/2	40.9	18.5	13/15	1.10	2.16	99.52
2/5	40.9	39.6	13/9	1.21	2.76	99.50
2/10	41.0	78.7	13/5	1.29	3.92	99.46
10/2	102.4	18.5	11/15	0.94	4.08	99.45
10/10	102.8	78.6	11/5	1.20	5.86	99.38

a few percent. The turn-off energy is almost independent of current: the gate is discharged through about $0.8\ \Omega$ faster than the load current can move the output charge, so the channel is off before V_{DS} rises, the voltage rises at $dv/dt \approx I/C_{oss,tot}$ (5.3 nF), and most of the 7–8 μJ measured at the device terminals is energy stored in C_{oss} .

A motor-friendly design may slow the switching with larger gate resistors. Fig. 5 and Table III compare six resistor pairs simulated with the full building-block model. The turn-off becomes channel-controlled only when the gate removes the Miller charge more slowly than the load current moves the output charge, i.e. for

$$R_{g,off,tot} > R_{crit} = \frac{V_{pl} Q_{oss,tot}}{I Q_{GD}}, \quad (2)$$

which gives about $1.5\ \Omega$ at 139 A and $7\ \Omega$ at 30 A ($V_{pl} \approx 2\ V$, $Q_{GD} = 3.8\ nC$). Above R_{crit} the voltage term of the classical overlap formula predicts E_{off} within about 10%, whereas the additional current-fall term $I^2 V / (2 di/dt)$ overestimates it by a factor of 1.8; below R_{crit} the formula overestimates the loss by a factor of two. At turn-on, the di/dt term is needed only when the gate resistance makes the current rise a separate, slow phase (10 Ω).

Two consequences follow for the drive design. First, $R_{g,on} = 10\ \Omega$ mainly slows the current rise: the turn-on dv/dt falls only from 13 to 11 V/ns while E_{on} rises 2.5 times; the turn-off dv/dt is controlled by $R_{g,off}$ only above R_{crit} . Second, the same $R_{g,off}$ holds the complementary transistor off, and its gate bump rises from 0.99 to 1.29 V; slowing the turn-off therefore requires a separate low-impedance hold-off path. The fully slowed 10/10 Ω design misses the efficiency target at 50 kHz.

V. LOSSES INCLUDING FREQUENCY-DEPENDENT PCB COPPER

The leg loss is evaluated for a sinusoidal current with continuous pulse-width modulation, for which conduction, switching and dead-time losses are identical for sinusoidal and space-vector modulation. With synchronous conduction, the high- and low-side transistors also share the loss equally for any modulation index and power factor, since $d\hat{i}^2 = \hat{i}^2/2$.

The PCB copper loss is usually computed with dc resistances. The dc-link current of a phase leg, however, contains about $29\ A_{rms}$ at the switching frequency and its harmonics, which circulates between the MLCC bank and the cells. A

TABLE IV
MAXIMUM TEMPERATURES (ICEPAK, 60 °C COLD PLATE)

	top cooling	top + bottom
EPC2361 junction	72 °C	67 °C
Board	104 °C	83 °C
1210 capacitors	103 °C	83 °C
Driver	91 °C	75 °C

Q3D frequency sweep from 1 kHz to 100 MHz shows that the branch resistances are already 1.5 times their dc value at 50 kHz and 3 times at 500 kHz (Fig. 6a), because the current crowds onto the facing plane surfaces and the lowest-inductance paths. The board is therefore solved as a frequency-domain network of the Q3D $R(f)$ and $L(f)$ matrices, the capacitor groups and the dc-supply impedance, excited by the switched transistor currents at every harmonic of the modulated waveforms:

$$P_{Cu} = \sum_h \frac{1}{2} \mathbf{I}_h^H \mathbf{R}(f_h) \mathbf{I}_h. \quad (3)$$

The load-frequency part reproduces the dc estimate (3.30 W); the switching harmonics add 1.07 W at the rated point and up to 2.5 W at low modulation index, where the ripple current is largest. The same network gives an MLCC loss of 0.39 W.

Fig. 6(b) and (c) show the resulting loss breakdown. At 50 kHz the leg dissipates 12.49 W (99.53%): 5.55 W conduction, 4.37 W copper, 1.91 W switching, 0.20 W dead time and 0.39 W in the capacitors. Conduction and copper account for about 80% of the loss, so cooling and copper design matter more than switching speed at this frequency, which is why the gate resistors could be selected for robustness. The 99.5% target holds up to about 60 kHz; at 100 kHz the efficiency is 99.45%, and with the maximum on-resistance the target is missed already at 50 kHz (99.46%). Without the switching-harmonic copper term, the same design would have appeared to reach 99.50% at 100 kHz.

The dc link of $24 \times 10\ \mu F$ 1210 and $12 \times 1\ \mu F$ 0805 capacitors (77 μF effective at 75 V) holds the ripple within $\pm 2\%$ when three legs share a submodule dc link at 50 kHz. Above about 98 kHz the number of capacitors is set by their ripple-current rating rather than by the voltage ripple, which bounds the useful switching frequency from the density point of view.

VI. THERMAL DESIGN

A steady-state conduction model is built in Ansys Icepak from the same geometry. The board is an anisotropic equivalent of FR-4 and copper (in-plane 55 W/mK, through-plane 0.36 W/mK, 3.5 W/mK in the cell regions with via arrays); each transistor is a 1.9 W source with a junction-to-case resistance of 0.2 K/W to the top and a junction-to-board resistance of 1.5 K/W; a 0.5 mm gap pad (1.9 K/W per device) and aluminium pedestals connect the transistors to a 3 mm aluminium plate held at 60 °C (liquid cold plate). The board copper loss (4.4 W), the capacitor losses and the driver loss are applied as volumetric sources.

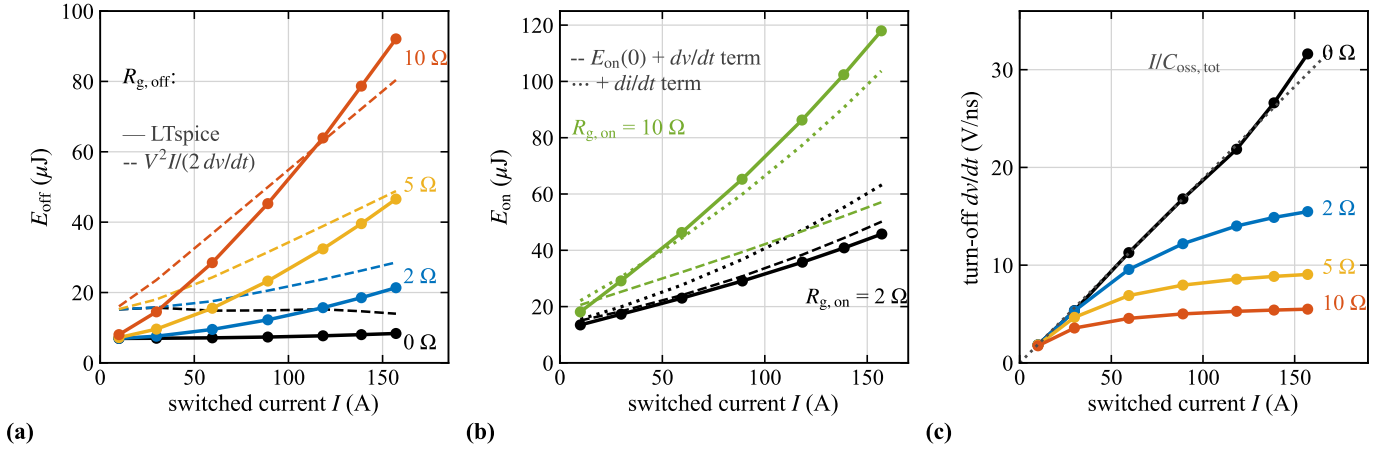


Fig. 5. Switching energies per switch position of the building block with slow switching (LTspice with the Q3D parasitics, 75 V, $R_{g,\text{on}}$ and $R_{g,\text{off}}$ external resistances). (a) Turn-off energy (solid) and voltage-overlap estimate $V^2 I / (2 dv/dt)$ with the simulated dv/dt (dashed) for $R_{g,\text{off}} = 0, 2, 5$ and 10Ω ; the estimate fails below the critical resistance (2). (b) Turn-on energy for $R_{g,\text{on}} = 2$ and 10Ω against the current-independent part $E_{\text{on}}(0)$ plus the voltage-overlap term (dashed) and plus the current-overlap term (dotted). (c) Turn-off dv/dt against the load-driven limit $I/C_{\text{oss,tot}}$ (dotted); without external resistance the voltage rise is set by the load current, with 5–10 Ω by the gate.

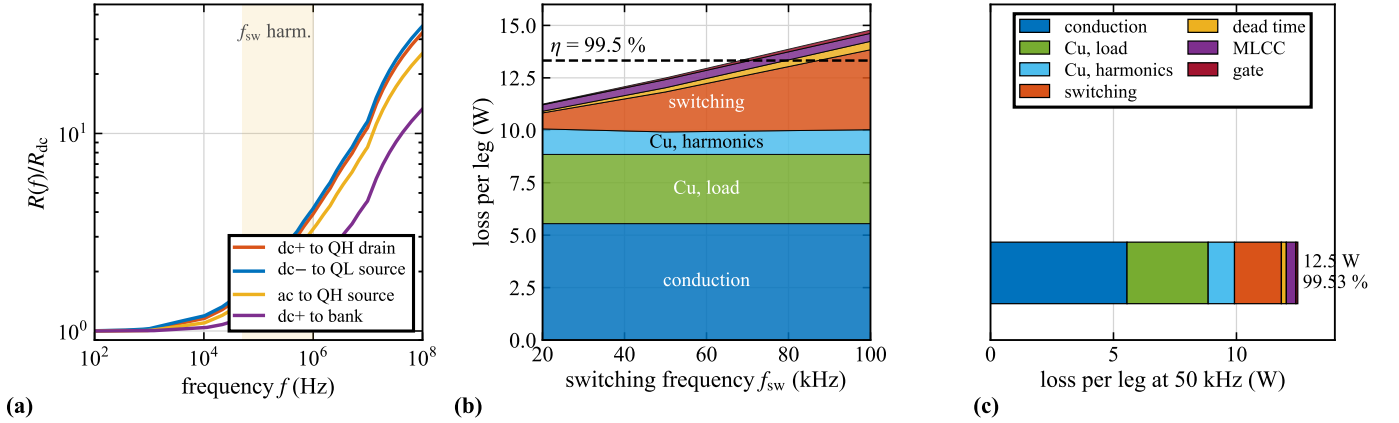


Fig. 6. Losses of the building block at the rated point (75 V, 100 A_{rms}, $M = 1$, typical $R_{\text{DS(on)}}$ at 100 °C). (a) Q3D resistance of the main board branches against frequency, normalized to the dc value; the shaded band covers the switching frequency and its first harmonics. (b) Loss per leg against switching frequency with the efficiency target. (c) Loss breakdown at 50 kHz; conduction and copper loss together account for about 80 %.

With the cold plate on the transistors only, the junctions stay at 72 °C (Table IV), 7 K above a per-device estimate, because the transistors also conduct the heat of the board. The board itself, however, reaches 104 °C in the capacitor bank (Fig. 7): its copper loss, larger than the loss of each transistor, has no path to the cold plate other than laterally through the board and upwards through the transistors. A bottom-side gap pad to a second plate lowers the board maximum to 83 °C and the junctions to 67 °C. Since the model distributes the copper loss over the board thickness and neglects convection, the absolute board temperatures are conservative; the conclusion that a low-voltage, high-current building block needs a cooling path for its board copper is not.

VII. DISCUSSION AND FUTURE WORK

Compared with the reviewed paralleled-GaN half bridges (Table V), whose commutation loops range from 0.34 to 6.4 nH, and up to 11 nH in a commercial module [11], the

TABLE V
COMPARISON WITH PARALLELED-GAN HALF BRIDGES

Work	Devices/switch	Rating / test	L_{loop}	Layout
[12]	4 × 650 V	400 V/240 A DPT	0.7 nH	6-layer PCB
[13]	4 × 100 V	48 V/328 A DPT	1.14 nH	IMS hybrid
[16]	8 × 200 V	120 V/155 A, 100 kHz	0.34 nH*	PCB, symm. gate
[15]	8 × 100 V	48 V/125 A _{rms}	—	butterfly PCB
[19]	4 × 650 V	350 A _{rms} WLTP	3.9 nH	PCB module
[17]	3 × 650 V	400 V, 100 kW	6.4 nH	PCB + Cu blocks
This work	2 × 100 V	75 V/141 A (design)	0.37 nH/cell	mirrored cells

*analytical estimate

proposed block reaches 0.374 nH per cell with two loops in parallel, and its symmetric gate drive removes the dominant sharing mechanisms. The analysis identifies four limits that guide the next iteration: the off-state gate bump, which calls for a Miller clamp, a small negative off-voltage or a gate-source capacitor [15]; the copper loss, which can be reduced

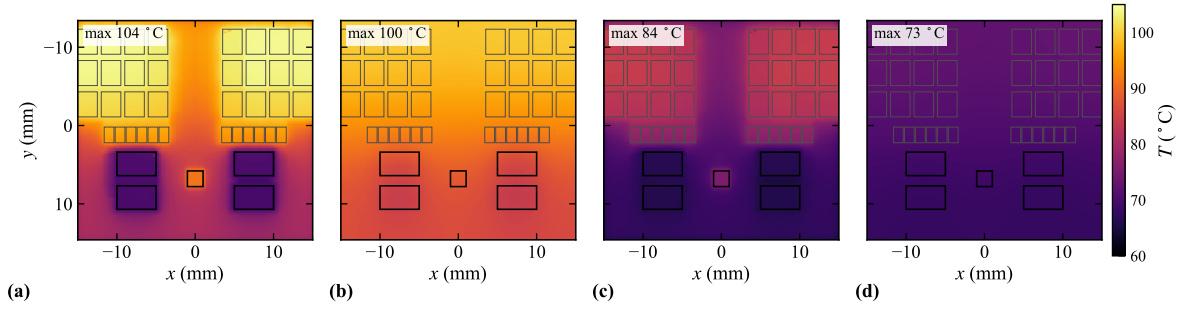


Fig. 7. Icepak temperature maps at the rated point (50 kHz) in the device layer ($z = 2.0$ mm, including the air between components) and the board mid-plane. (a), (b) Cold plate on the transistors only. (c), (d) Additional bottom-side gap pad (3 W/mK) to a second plate at 60 °C. Outlines: transistors, MLCCs and driver.

by about 2 W with busbar contacts next to each cell and heavier inner layers, and further by placing more capacitance next to the cells to shorten the ripple path; the thermal path of the board; and the dc-link ripple on a shared submodule bus, which is at its limit.

All results in this paper are from analysis, finite-element extraction and circuit simulation. Experimental validation is future work and will proceed from double-pulse tests at 75 V and up to 160 A, including the loop inductance from the ringing frequency (260–330 MHz expected) and the per-transistor currents, to continuous operation for efficiency and thermal-image based sharing checks [16], and finally to an SPB submodule driving a multiphase machine, where the trade-off between inverter and machine losses will set the final switching frequency [21].

VIII. CONCLUSION

A building block for stacked polyphase bridge converters has been designed for 75 V and 100 A_{rms} with two paralleled 100 V GaN transistors per switch. Mirrored commutation cells with vertical loops and a centre gate driver give 0.374 nH per cell, equal 1.7 nH gate loops and a predicted dynamic current imbalance below 0.1 %. At 50 kHz the block reaches 99.53 % efficiency at 2.65 kW per leg on 8.5 cm². The design shows that the switching harmonics of the dc-link current add about one third to the dc copper loss of the board, that the overlap loss formula applies to a GaN turn-off only above the critical gate resistance (2), and that the board copper, rather than the transistors, sets the thermal design of a low-voltage, high-current building block.

REFERENCES

- [1] S. Norrga, L. Jin, O. Wallmark, A. Mayer, and K. Ilves, "A novel inverter topology for compact EV and HEV drive systems," in *Proc. IECON*, 2013, pp. 6590–6595.
- [2] L. Jin, S. Norrga, O. Wallmark, and M. Nikouei Harnefors, "Control and modulation of the stacked polyphase bridges inverter," in *Proc. IEEE ECCE*, 2014.
- [3] L. Jin, S. Norrga, O. Wallmark, and N. Apostolopoulos, "Modulation and power losses of a stacked polyphase bridge converter," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 5, no. 1, pp. 409–418, Mar. 2017.
- [4] L. Jin, S. Norrga, and O. Wallmark, "Analysis of power losses in power MOSFET based stacked polyphase bridges converters," in *Proc. IPEDC-ECCE Asia*, 2016.
- [5] M. Nikouei, O. Wallmark, L. Jin, L. Harnefors, and H.-P. Nee, "DC-link stability analysis and controller design for the stacked polyphase bridges converter," *IEEE Trans. Power Electron.*, vol. 32, no. 2, pp. 1666–1674, Feb. 2017.
- [6] M. Nikouei, H. Zhang, O. Wallmark, and H.-P. Nee, "A highly integrated electric drive system for tomorrow's EVs and HEVs," in *Proc. IEEE SPEC*, 2017.
- [7] H. Zhang, O. Wallmark, M. Leksell, S. Norrga, M. Nikouei Harnefors, and L. Jin, "Machine design considerations for an MHF/SPB-converter based electric drive," in *Proc. IECON*, 2014.
- [8] T. Bringezu and J. Biela, "Comparison of optimized motor-inverter systems using a stacked polyphase bridge converter combined with a 3-, 6-, 9-, or 12-phase PMSM," in *Proc. EPE ECCE Europe*, 2020.
- [9] T. Bringezu and J. Biela, "Control of a 9-phase PMSM with stacked polyphase bridge converter including dc source impedance," in *Proc. EPE ECCE Europe*, 2023.
- [10] L. Verkroost, F. De Belie, L. Faria da Rocha, P. Keim Olsen, P. Sergeant, and H. Vansompel, "Multi-agent control scheme for power distribution in a multi-three-phase motor drive fed by a stacked polyphase bridges converter," *IEEE Trans. Energy Convers.*, vol. 39, no. 4, pp. 2568–2580, Dec. 2024.
- [11] J. A. Brothers and T. Beechner, "GaN module design recommendations based on the analysis of a commercial 3-phase GaN module," in *Proc. IEEE ECCE*, 2019, pp. 4109–4116.
- [12] J. Lu and D. Chen, "Paralleling GaN E-HEMTs in 10 kW–100 kW systems," in *Proc. IEEE APEC*, 2017, pp. 3049–3056.
- [13] M. T. Tran *et al.*, "A high-performance GaN power module with parallel packaging for high-current and low-voltage traction inverter applications," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 13, no. 1, pp. 1188–1209, Feb. 2025.
- [14] S. Musumeci, V. Barba, and M. Palma, "GaN-based low-voltage inverter for electric scooter drive system," in *Proc. AEIT Int. Annu. Conf.*, 2022.
- [15] M. Wattenberg, O. G. Lorenz, and J. Sanchez, "A multi-kilowatt low-profile GaN inverter for light electric vehicles and high-power tools," in *Proc. IEEE APEC*, 2023, pp. 1443–1450.
- [16] M. W. J. Cooke and D. J. Rogers, "120 V, 155 A traction inverter using eight parallel GaN HEMTs," 2026.
- [17] M. Zou *et al.*, "Systematic efficiency-density co-optimization of 100 kW GaN traction inverter: methodology and integration," *IEEE Trans. Power Electron.*, vol. 40, no. 11, pp. 16281–16299, Nov. 2025.
- [18] M. D. Rahman *et al.*, "Design and evaluation of a 650 V, 300 A GaN-based power module with integrated drivers and ultra-low inductance layout," in *Proc. IEEE WiPDA*, 2025.
- [19] D. Novo *et al.*, "Development and testing of GaN power modules for EV traction inverters," in *Proc. CIPS*, 2024, pp. 692–698.
- [20] G. Rohner, J. Huber, S. Mirić, and J. W. Kolar, "Comparative evaluation of three-phase three-level flying capacitor and stacked polyphase bridge GaN inverter systems for integrated motor drives," *Electronics*, vol. 13, no. 7, Art. no. 1259, 2024, doi: 10.3390/electronics13071259.
- [21] M. Cooke and D. J. Rogers, "Effect of switching frequency on efficiency in wide bandgap based drive systems," in *Proc. IEEE WiPDA*, 2024.